Google Cloud H2S

AI Builder Cup

Build in JAPAC.
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Team OpsCrew

Expense exception plans with human confirmation

Captain: Sunday Lee
Theme: Future of Work & Enterprise Productivity

Competition proposal / 16 October 2026

Brief about the idea

OpsCrew helps operations staff turn a synthetic expense exception into a supported handling plan.

Gemini extracts facts, proposes steps, then reviews the proposal in a separate call. Fixed policy checks constrain each stage.

A person reviews the current plan and explicitly confirms a case. The app reads the saved record independently before showing success.

The prototype covers one synthetic USD policy. It creates handling records and performs no reimbursement or payment.

Opportunities and proposed value

Problem: staff must reconcile intake text, missing receipts and policy thresholds before deciding what to request next.

Proposed difference: OpsCrew binds the plan to supported facts and exact policy clauses, then stores the person's confirmation with that version.

Design value: input edits invalidate old plans. Contradictions and unsupported citations block case creation.

Expected impact: clearer exception triage and fewer unsupported actions. Actual staff time savings and customer adoption remain unmeasured.

Features offered by the solution

Synthetic presets and original text citations make intake decisions inspectable.

Facts, Planner and Reviewer outputs expose disagreements and exact policy evidence.

Firebase visitor identity, owned history and version-bound confirmation protect each person's handling records.

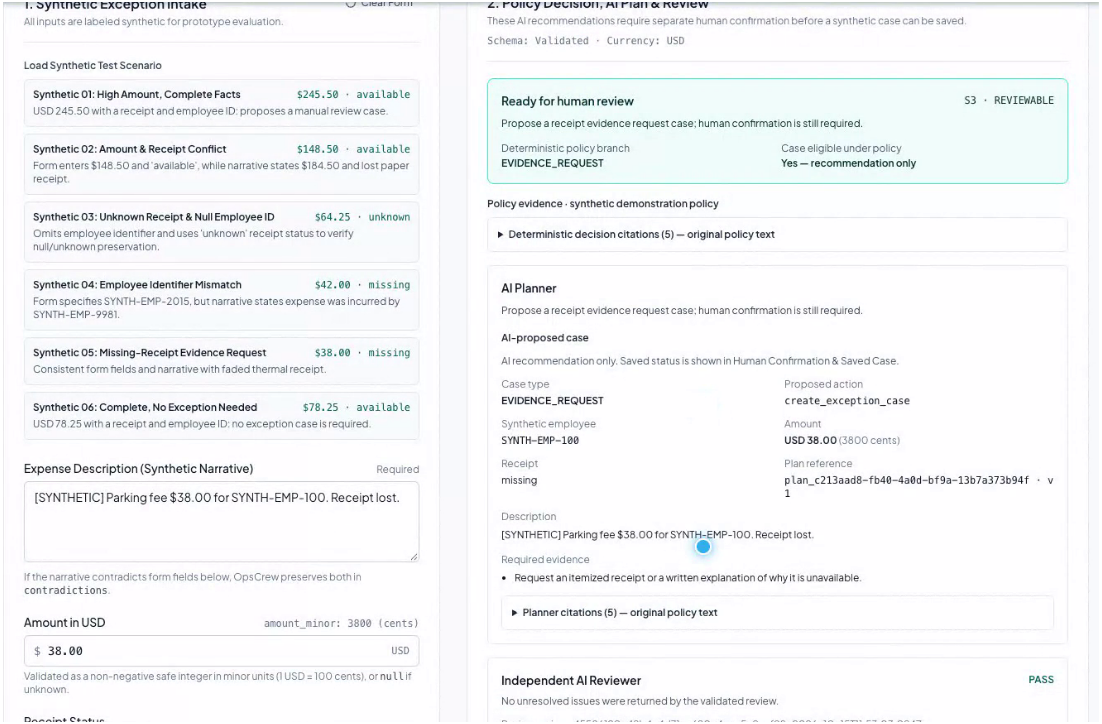
Atomic writes, idempotent recovery and independent readback prevent duplicate or unverified success. Shared quotas bound model traffic.

Process flow and confirmation



Missing evidence produces a request plan. Contradictions block progression.
Editing intake requires a new analysis and a new confirmation.

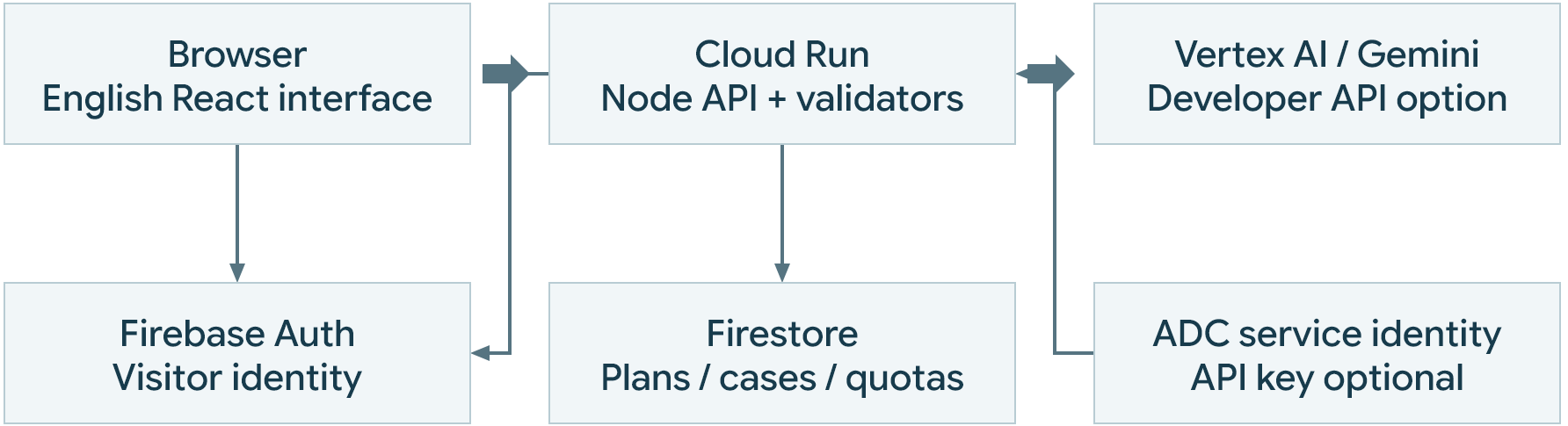
Interface in the recorded cloud demonstration



A person reviews the current result before confirming an eligible handling record.

Recorded UI, 15 October 2026, at 75 s. Synthetic example supplied by the team.

Cloud architecture and trust boundaries



The server selects Vertex AI with ADC or the optional Developer API route. Model credentials stay on the server. Visitor identity controls access to plans and saved cases.

Technologies used in the solution

Component	Role
Google AI Studio Build	Fresh application origin on 7 October
Gemini server modes	Vertex AI with ADC; optional Developer API
Current service config	Gemini 2.5 Flash, self-reported on 16 October
Node 24 / React / TypeScript	Explicit confirmation and fixed API policy checks
Cloud Run + Firebase	Hosting, visitor identity and saved handling records

From zero in AI Studio Build. Antigravity assisted deployment (team-reported).

Implementation cost and traffic controls

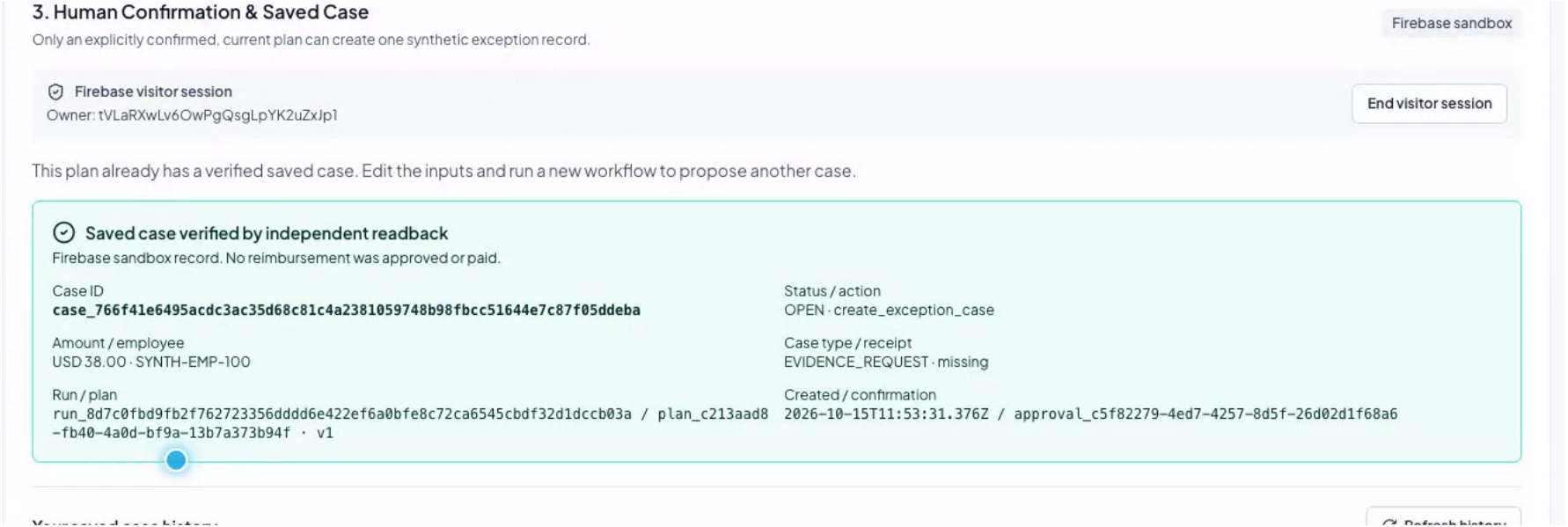
Team management target: USD 40. Billing alerts and application quotas cannot guarantee a monetary spending cap.

Default model limits: 2 concurrent runs globally, 1 per visitor. Each visitor has 2 runs per minute and 12 per UTC day.

Global limits: 6 runs per minute and 60 per UTC day. Each workflow reserves up to 6 provider attempts. Unknown usage receives no refund.

Historical 13 October Developer API observations: 10 attempts and 11,680 known tokens. One FACTS call returned 503. Its usage and actual billed cost are unknown.

Prototype: confirmed case and independent readback



Recorded UI, 15 October 2026, at 96 s. The confirmed USD 38 synthetic request is OPEN. No reimbursement approval or payment occurs.

Performance report and evidence boundaries

Evidence	Result	Method
Historical cloud core	11 / 11	3.1 Flash-Lite Developer API, 13 Oct
Historical cloud safety	14 / 14	13 Oct, includes overlapping core checks
Historical real workflows	3 public cases	3.1 Flash-Lite, 13 Oct
Offline engineering	246 / 246	2026-10-16 local regressions
Native artifact checks	12 groups	9 HTTP checks included, 2026-10-16

13 October workflow observations: 8.381-12.042 s, including recovery from one 503. These results do not certify the current Vertex backend, an SLA or independent held-out acceptance.

Additional details and future development

Extend deployment operations checks and provider regression coverage. The retained real-model API evidence is dated 13 October; current provider switching has separate offline tests.

Current evidence uses public synthetic examples. Independent held-out evaluation and staff timing studies remain future validation work.

The demonstration presents recorded interface operation with English narration, synthetic data and disclosed edits. A person reviews each eligible plan before creating its case.

After evaluation, consider broader policies and staff timing studies. Production reimbursement integrations require a separate design and authorization.

Repository, demo and final product links

Public GitHub repository

github.com/sundaylee3100-ljl/AI-Builder-Cup-2026-OpsCrew

Cloud product URL

https://opscrew-883881551915.us-central1.run.app

Demo video

https://youtu.be/j94ffx_yrng

Created in Google AI Studio Build on 7 October. Historical cloud API evidence: 13 October. Integrated offline checks: 2026-10-16.